BMC Plant Biology

RESEARCH Open Access

Identification of QTLs for aphid (Melanaphis sacchari) resistance in sorghum (Sorghum bicolor) based on BSA-seq and analysis of candidate genes

Junli Du[1,3†], Kunliang Zou[1,3†], Qi Shen[1,3], Yang Liu[2], Minghui Guan[2,3], Tonghan Wang[1], Degong Wu[1], Lihua Wang[1,3], Yongfei Wang[1,3] and Jieqin Li[1,3*]

Abstract
Aphid infestation significantly reduces the yield and forage quality of sorghum, underscoring the urgent need to identify and develop aphid-resistant sorghum varieties. Marker-assisted selection (MAS) remains one of the most effective strategies for enhancing aphid resistance in sorghum. In this study, an F_2 segregating population derived from a cross between the aphid-resistant variety IS23514 and the susceptible variety IS23992 was employed to map aphid resistance genes using bulked segregant analysis based on whole-genome resequencing (BSA-seq) and Insertion-Deletion (InDel) marker analysis. Genetic analysis indicated that aphid resistance in sorghum is governed by a dominant gene. A quantitative trait locus (QTL) associated with aphid resistance (qMsa-6) was identified within a 124 kb region on chromosome 6. Candidate gene prediction and protein domain analysis within this region revealed ten previously uncharacterized genes, seven of which possess NB-ARC domains. Subsequent qPCR analysis revealed that the SbiRTX2783.06G018400 gene is the most likely candidate conferring aphid resistance. This study offers critical molecular insights into the mechanisms underlying aphid resistance in sorghum and identifies key candidate genes for future aphid-resistant breeding programs.

Keywords Sorghum bicolor, Melanaphis sacchari, BSA-seq, InDel, QTL

Introduction

Sorghum (*Sorghum bicolor*,2n = 2x = 20), belonging to the Poaceae family, has a long cultivation history, with its wild ancestors dating back 8,000 years. It was first domesticated in Ethiopia and Sudan [1]. As the fifth largest grain crop globally, sorghum is primarily used for human consumption, silage feed, and as important materials for industrial and energy development [2]. Its broad agricultural climate adaptability and high yield characteristics are gradually expanding its cultivation area, enabling it to spread from the Sahel (the region between the Sahara Desert and the central Sudan savannah) to temperate and

†Junli Du and Kunliang Zou contributed equally to this work.

*Correspondence:
Jieqin Li
wlhljq@163.com
[1]College of Agriculture, Anhui Science and Technology University, Chuzhou 233100, China
[2]College of Resources and Environment, Anhui Science and Technology University, Chuzhou 233100, China
[3]Anhui Province International Joint Research Center of Forage Bio-breeding, Chuzhou 233100, China

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subtropical regions in Asia and the Americas [3]. Furthermore, sorghum boasts considerable genetic diversity, offering abundant resources for disease and pest resistance from both local cultivated varieties and their wild relatives [4].

The sorghum aphid (*Melanaphis sacchari*) is a significant pest affecting sorghum across Asia, Africa, Australia, and South America. It primarily feeds by piercing the undersides of sorghum leaves and stems, extracting plant sap, which leads to nutrient, sugar, and water deficiency, causing stunted growth, yellowing, wilting, or even the death of the plant. Furthermore, aphid infestation can significantly damage the forage quality of sorghum, resulting in reduced milk yield and quality in dairy cattle and even affecting the operation of agricultural machinery [5–7]. Insects contribute to substantial crop losses, estimated at up to 40% of global production annually, resulting in an economic loss of $470 billion [8]. While insecticides remain the primary control method, their long-term use may lead to resistance issues. Studies have shown that frequent application of insecticides can promote the evolution of resistance genes in aphids, making them resistant to these chemicals [9]. Therefore, utilizing modern biotechnologies such as MAS and gene editing to accelerate the identification and introduction of aphid resistance traits has become a research hotspot in sorghum breeding [10].

Aphid damage significantly impacts both the yield and stability of sorghum. Resistance to aphids in sorghum is currently categorized into four types: non-resistance, antibiosis, tolerance, and avoidance [11]. Among these, Resistance (R) genes play a crucial role in the aphid resistance response of sorghum. They function by recognizing aphid effectors and activating immune responses, thereby enhancing the plant's resistance and tolerance. R genes provide specific recognition of pathogen effectors and trigger defensive reactions, playing a key role in the plant's innate immune system. Typically, their proteins consist of three conserved domains: the Nucleotide-binding site (NBS) domain, which is involved in ATP or GTP binding and hydrolysis; the Leucine-rich repeat (LRR) domain, responsible for effector recognition; and the NB-ARC domain, which acts as a molecular switch regulating the activation of downstream signaling pathways [12, 13]. However, the rapid evolution of pathogens often undermines the resistance provided by R genes, creating significant challenges for their application in crops. This ongoing "arms race" between plant hosts and pathogens underscores the urgent need for the continual discovery and functional characterization of new R genes, as well as the development of additional markers to enhance strategies for achieving durable resistance [14]. With the advancements in high-throughput sequencing technologies, genomics breeding, and artificial intelligence have significantly expedited the development of aphid-resistant varieties. Extensive studies on the molecular mechanisms governing plant-aphid interactions, through QTL screening and molecular mapping, have made molecular MAS an effective approach for breeding aphid resistance [15]. Sorghum BTx623 is the most widely used genome today, having undergone multiple updates and assemblies since its release [16, 17]. This genome has been used to locate aphid resistance genes using Simple sequence repeat (SSR) markers, identifying the aphid resistance gene *RMES1* in the Chinese sorghum variety HN16, which lays the foundation for the genetic improvement of aphid resistance in sorghum [18]. Using the *RMES1* aphid resistance gene, researchers developed Sequence characterized amplified region (SCAR) markers and successfully identified a high-quality aphid-resistant sweet sorghum variety (BSS507) [19]. Muleta et al. analyzed the proximity of the *RMES1* locus and its flanking sequences for the applicability of Kompetitive allele specific PCR (KASP) markers, developing the sbv3.1 1_06_02892438r KASP marker, This marker has demonstrated reliability and predictability for use in sorghum resistance breeding. Validation by commercial companies indicated that this marker aligned with over 80% of prevalent planting resources, showing a matching rate higher than 99.5% in technical repeat tests [20]. This research not only sheds light on the genetic foundations of aphid resistance traits in sorghum but also provides valuable candidate gene markers for molecular marker-assisted breeding. In the future, the application of gene editing or molecular breeding techniques is anticipated to effectively incorporate aphid resistance traits into sorghum breeding programs, thereby improving pest and disease resistance while ensuring agricultural production stability and economic benefits.

Building on this foundation, utilizing molecular breeding techniques for comprehensive management of aphids appears to be an effective solution to this problem. To gain a more comprehensive understanding of sorghum's resistance to aphids and enhance its survival potential under aphid pressure, this study used two extreme parental breeding lines, sorghum IS23514 and sorghum IS23992, to successfully construct an F_2 segregating population containing the target gene. By combining BSA-seq with InDel markers, we identified a novel aphid resistance QTL within an approximately 124 kb region on sorghum chromosome 6 and predicted genes associated with aphid resistance.

Materials and methods

Screening of aphid-resistant plant materials and aphid rearing

The sorghum lines used in this study, IS23514 (aphid-resistant) and IS23992 (aphid-susceptible), were selected from the core sorghum germplasm collection at Anhui Science and Technology University based on preliminary aphid resistance evaluation. The resistance levels were determined by measuring aphid population sizes after artificial infestation under controlled conditions (see Methods section for details). The plants were cultivated in an artificial climate chamber under a 16/8 h light/dark cycle, at 28 °C with 50–60% relative humidity. They were watered and fertilized regularly. Seedlings were grown in plastic pots (7 cm height × 6 cm width, 0.5 L volume) and enclosed with mesh to prevent aphid escape. Aphids used for infestation were originally collected from the sorghum cultivar "Tiegan Kangsi" and transferred to fresh plants every two weeks to maintain a stable colony.

Evaluation of sorghum damage

At the three-leaf stage of sorghum growth, 10 third- or fourth-instar apterous aphids were transferred onto the sorghum plants using a fine brush. During the aphid infestation, the number of aphids was recorded daily for one week. After seven days of infestation, the damage to the sorghum plants was assessed, and aphid resistance levels were categorized based on the maximum aphid count. The method, slightly modified from J. Zhang et al. [18]. classified sorghum into five aphid resistance levels (Table 1).

Construction of the sorghum segregating population

Based on aphid resistance screening, the highly aphid-resistant sorghum (IS23514) and aphid-susceptible sorghum (IS23992) were selected as parental lines. These lines were crossed in the experimental fields of Anhui University of Science and Technology, resulting in the F_1 generation. Subsequently, the F_1 generation was cultivated at the sorghum breeding base in Hainan, where self-pollination was performed to develop an F_2 segregating population (See Supplementary Fig. 1). Both the F_1 and F_2 populations were randomly planted in 4×8 plug trays. When the sorghum plants reached the three-leaf stage, 10 third- or fourth-instar apterous aphids were introduced to each plant. Aphid numbers were continuously monitored over seven days, and the data were compared to those of the parental lines. Finally, statistical analyses of aphid counts on sorghum plants were conducted using Excel 2021 and GraphPad Prism 10.1.2, through which the dominance relationship and inheritance pattern of the aphid resistance gene were determined.

Table 1 Classification of aphid resistance levels in sorghum

Aphid Quantity	Resistance Level
$X < 20$	Highly Resistant
$20 \leq X < 30$	Resistant
$30 \leq X < 60$	Intermediate
$60 \leq X < 80$	Susceptible
$X \geq 80$	Highly Susceptible

DNA extraction and BSA sequencing processing of sorghum

In this study, four DNA pools were constructed based on aphid resistance assessments: two parental pools (IS23514 and IS23992) and two extreme phenotype pools, derived from the F_2 lines, representing high resistance and high susceptibility. DNA was extracted using the DNA Secure Plant Kit from TianGen Biochemical Technology (Beijing) Co., Ltd., achieving a concentration of 20 ng/μL. To ensure sufficient sequencing depth, at least 20 samples were combined in equal proportions to create the extreme mixed pools. Sequencing was conducted by BGI utilizing the HiSeq 2000 platform. The quality of the raw sequencing data was evaluated using FastQC software to ensure high-quality genomic sequences for each pool. Contaminated and low-complexity sequences were subsequently removed using the BBduk tool from the BBMap suite (v38.90). For alignment, BWA software was employed to map the sequences to the sorghum reference genome (Sorghum bicolor v3.1.1). SNPs were identified and filtered using SAMtools to calculate the SNP index for both high and low phenotype pools. The ΔSNP index for each SNP was computed to illustrate the differences between the two phenotype pools. A sliding window approach, employing a 1 Mb window size and a 200 kb step size, was utilized to smooth the ΔSNP index data, generating a comprehensive ΔSNP index map of the entire genome to pinpoint potential QTL regions. Regions with consistently elevated ΔSNP indices were deemed candidate QTLs.

Primer design, screening, and indel molecular markers

By comparing the sequenced sorghum genome to identify linked regions, the analysis began with a 10 Mb interval, which was gradually narrowed as the distance to the target gene decreased. Sequences of microsatellite markers extending 200 bp on either side were downloaded from the Phytozome database (https://phytozome-next.jgi.doe.gov/) for the sorghum genome (Sorghum bicolor v3.1.1) [21]. Primers for the microsatellite sequences were designed using Primer 6 software and synthesized by Nanjing Genscript Biotech Corporation. A 10% Polyacrylamide Gel Electrophoresis (PAGE) was employed to verify the polymorphism differences between the parents (IS23514 and IS23992) for screening purposes. InDel molecular markers that showed differences were used to perform PCR amplification on the homozygous individuals of the F_2 segregating population, further narrowing down the target gene region.

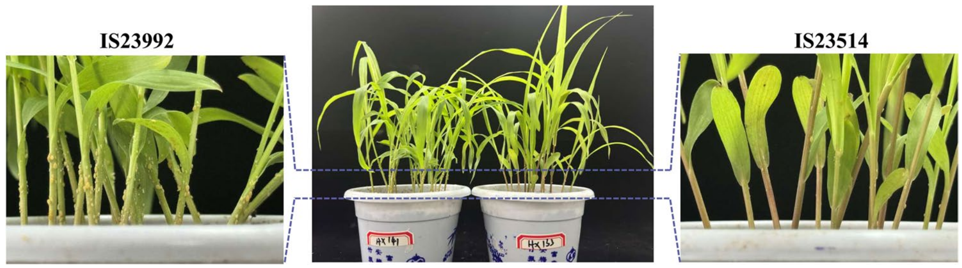

Fig. 1 Differences in Resistant and Susceptible Sorghum Parents After Aphid Infestation

Construction of the linkage map

InDel molecular markers that exhibited differences were used to perform PCR amplification on the homozygous individuals of the F_2 segregating population. The amplified products were then separated by PAGE electrophoresis. Individuals displaying banding patterns consistent with the aphid-resistant parent IS23514 were classified as maternal-type, while those matching the aphid-susceptible parent IS23992 were classified as paternal-type. Progeny exhibiting both banding patterns were designated as heterozygous. Statistical analysis of the genotypes was conducted, and the map function of QTL IciMapping 4.2 software was used to construct the genetic linkage map [22].

Analysis of candidate genes

To identify candidate genes, the sequences from the resistance sorghum IS23514 and the susceptible sorghum IS23992 were mapped to the identified resistant sorghum RTx2783 and BTx623 [23, 24]. Bioinformatics analysis was performed on all candidate genes within the target gene interval. The protein domains of all target genes were predicted and classified using the InterPro online tool (https://www.ebi.ac.uk/interpro/). Subsequently [25], motifs were predicted using the MEME online tool (https://meme-suite.org/meme/tools/meme) with the motif count set to 9 and other default settings. Finally, the results were visualized using Tbtools [26, 27].

qPCR analysis of candidate resistance gene expression

To investigate the function of candidate genes, samples were collected at various time points following aphid infestation, including early (6 h, 24 h, 48 h) and late (7d) stages. Control leaf samples were also taken from uninfected plants at 0 h and 7d. RNA was extracted from the leaves, following standard methods [28]. qPCR primers were designed with Premier 6 software, referencing the endogenous control genes outlined by J. Li et al. [29], and primer specificity was confirmed using Primer-BLAST [30]. The qPCR experiments were conducted according to the method by Jin et al. [31], utilizing the ABI ViiA 7 (Life Technologies, Carlsbad, CA, USA) for detection. Data analysis was performed using the $2^{-\Delta\Delta Ct}$ method [32]. Statistical analysis and data visualization were conducted with IBM SPSS Statistics 24 and GraphPad Prism 8 software. Each qPCR reaction was conducted in triplicate to ensure the reliability of the results.

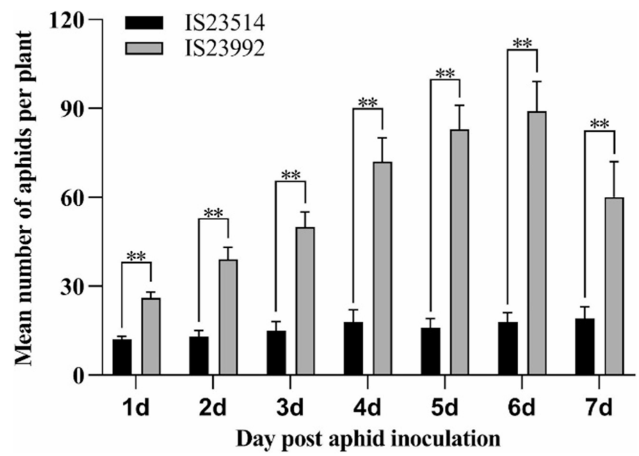

Fig. 2 Aphid Counts on Resistant and Susceptible Sorghum Plants After Aphid Infestation. Significant differences are indicated by (** = $p < 0.01$, * $= p < 0.05$)

Results and analysis

Analysis of aphid resistance assessment results

In this study, two sorghum varieties with distinct differences in aphid resistance were identified (Fig. 1): a highly resistant sorghum variety IS23514 and a highly susceptible sorghum variety IS23992. The results showed that the aphid population on IS23514 increased slowly over time, reaching a peak on day 4 before stabilizing. In contrast, the aphid population on IS23992 increased rapidly and continuously, peaking on day 6, followed by a gradual decline, primarily due to wilting and near death of the susceptible sorghum stems (Fig. 2). All experiments were repeated three times, and the results confirmed that the performance of IS23514 and IS23992 was consistent with expectations.

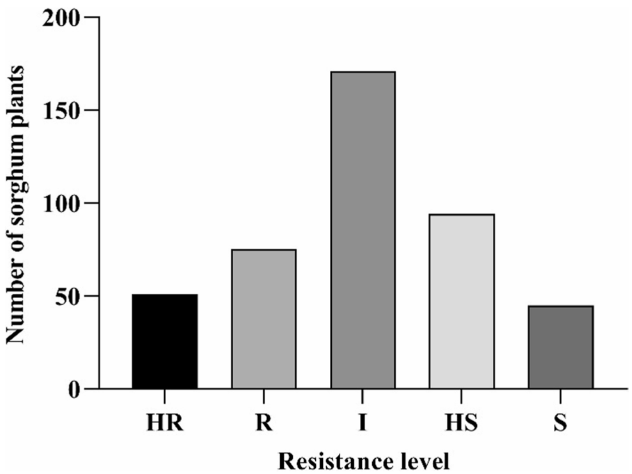

Fig. 3 Distribution of Plant Numbers with Different Resistance Levels in the Sorghum F_2 Population

Analysis of sorghum segregating population and resistance genes

A hybrid population was constructed using IS23514 as the maternal parent and IS23992 as the paternal parent, followed by aphid resistance identification in the F_1 and F_2 segregating populations. A total of 93 F_1 plants and 436 F_2 plants were randomly selected for classification and statistical analysis (See Supplementary Table 1). All F_1 plants were classified as either highly resistant or resistant, with an approximate ratio of 1:1, and no susceptible plants were observed. This suggests that the aphid resistance gene is dominant. In the F_2 population of 436 plants, the following distribution was observed: 51 highly resistant, 75 resistant, 171 intermediate, 94 susceptible, and 45 highly susceptible plants (Fig. 3). Overall, the segregation did not conform to Mendelian inheritance for a single gene, indicating that aphid resistance in sorghum is controlled by multiple genes.

Quality assessment of BSA-seq sequencing data

In this experiment, four BSA-seq databases were constructed, and after filtering and screening, the Clean Reads obtained from various samples ranged from 88.05 to 155.16 Mb, with Q20 values between 97.22% and 98.82% and GC content ranging from 42.82 to 43.25%. The average sequencing depth was between 31.77 and 38.44. Over 98.43% of the Clean Reads successfully mapped to the sorghum reference genome RTx623, with haploid coverage ranging from 91.62 to 93.49% (Table 2). These results indicate that the sequencing depth is sufficient, the alignment rates with the reference genome are high, and the haploid coverage is uniform, demonstrating good quality of the sequenced regions, suitable for subsequent experiments.

Analysis of BSA-seq sequencing results

To identify the locus of the aphid resistance gene, this study performed BSA-seq on four DNA pools: two parental pools (IS23514 and IS23992) and two extreme resistance pools (R-pool and S-pool). The association analysis of detected SNPs and Indels revealed that IS23514 had the lowest SNP count (3,288,597), while the S-pool had the highest (5,306,317). The R-pool and S-pool exhibited greater SNP counts than the individual parental samples, reflecting increased variability due to the combination of genetic material from multiple individuals. IS23514 had the lowest heterozygosity rate (HET rate) at 39.10%, suggesting reduced genetic variation, likely due to inbreeding within the sample. In contrast, IS23992 showed a significantly higher HET rate of 52.91%, indicating greater genetic diversity. Both pooled samples (R-pool at 70.55% and S-pool at 70.53%) exhibited high heterozygosity, with minimal differences from the F_2 population compared to the reference genome. Conversely, IS23514 displayed considerable divergence from the reference. The Ts/Tv ratio among all samples ranged from 1.90 to 1.97, indicating high-quality genomic data, as a ratio between 1.8 and 2.0 is associated with accurate SNP calling. Indel analysis highlighted distinct patterns of genetic variation between individuals and pooled samples. IS23514 exhibited low genetic diversity, with fewer heterozygous Indels and more homozygous variants, while IS23992 displayed greater diversity, featuring more heterozygous and fewer homozygous variants. The pooled samples showed the highest heterozygosity, consistent with the contribution of multiple genomes (Table 3). Finally, the BSA-seq linkage analysis indicated a significant positive area on chromosome 6, allowing for the localization of the aphid resistance gene within the 0–50 Mb range (Fig. 4).

InDel molecular marker analysis and validation

Based on the distribution of the ΔSNP-index across the sorghum chromosomes, the target gene was initially localized to the 0–50 Mb region on chromosome 6.

Table 2 Statistics of BSA-seq sequencing data

Sample	Clean Reads	Mapped Rate (%)	Q20(%)	GC Content (%)	Average Depth	Coverage Ratio 1× (%)
IS23514	155,160,588	98.56%	97.22%	43.25%	33.4337	91.6178
IS23992	150,234,788	98.44%	97.24%	43.23%	31.7710	93.1852
R-pool	88,051,148	98.80%	98.64%	42.99%	38.4388	93.4918
S-pool	88,155,279	98.43%	98.82%	42.82%	38.3131	93.4431

Table 3 Genotype statistics for each sample in the BSA-seq population

Sample	Total SNPs	HET SNPs	HOM_ALT SNPs	Het Rate	Ts/Tv
	Total Indels	**HET Indels**	**HOM_ALT Indels**	**Insertions**	**Deletions**
IS23514	3,288,597	1,285,932	1,997,427	39.10%	1.95
	464,880	112,339	350,803	133,443,506	28,766
IS23992	4,126,018	2,183,223	1,938,032	52.91%	1.91
	606,871	283,525	321,890	152,914,266	36,913
R-pool	5,329,411	3,759,677	1,566,695	70.55%	1.97
	815,343	593,062	221,322	95,773,696	52,814
S-pool	5,306,317	3,742,375	1,560,580	70.53%	1.97
	814,112	590,703	222,399	93,407,452	52,494

Within this region, we selected 234 pairs of InDel markers for PCR analysis to verify polymorphisms between the parents. Out of the 234 pairs of primers amplified, 42 pairs successfully revealed polymorphisms between the parents (Fig. 5). Subsequently, the selected 42 polymorphic primers were used to perform PCR amplification on 493 aphid-susceptible individuals from the F_2 population (See Supplementary Fig. 2). The amplification results were analyzed using PAGE electrophoresis to identify InDel polymorphic markers closely linked to the target gene. Ultimately, the target gene was localized between the left InDel marker Sb6M2672 and the right InDel marker Sb6M2796.

Construction of indel linkage map and analysis of candidate genes

Through InDel marker mapping, we identified multiple overlapping InDel markers in the region of chromosome 6 of sorghum. This overlapping region was selected as the final candidate gene region, located between the InDel markers SbM2672 and SbM2796 (See Supplementary Table 2). The physical distance is 2.672–2.796 Mb, with a region size of approximately 124 Kb and a genetic distance of 0.81 cm. Additionally, a new QTL was identified within this interval, named qMsa-6 (Fig. 6). Comparative analysis revealed that the resistant sorghum contained ten uncharacterized genes more than the susceptible sorghum in this region (Fig. 7). To investigate the functions of these ten genes, we performed protein domain and conserved motif analyses. It was found that seven of these genes (*SbiRTX2783.06G018200*, *SbiRTX2783.06G018300*, *SbiRTX2783.06G018400*, *SbiRTX2783.06G018450*, *SbiRTX2783.06G018500*, *SbiRTX2783.06G018800*, and *SbiRTX2783.06G018900*) contained a NB-ARC domain. The primary conserved motifs were identified as Motif2, Motif3, and Motif5 (Fig. 8). These domains, along with NB-LRR, NBS-LRR, and NLR, are collectively referred to as R genes. They can regulate downstream signaling pathways, activating defense responses in plants.

Expression analysis of candidate resistance genes by qPCR

To elucidate the role of aphid resistance candidate genes during aphid infestation in sorghum, we subjected the highly resistant variety IS23514 to aphid stress treatments and designed primers for seven candidate genes (See Supplementary Table 3) for qPCR analysis of gene expression. The results indicated (Fig. 9) that, at 6 h post-treatment, the relative expression levels of all candidate genes in the resistant variety IS23514 were higher than those in the control group. Notably, the *SbiRTX2783.06G018400* gene exhibited a strong induction, with expression levels significantly upregulated to approximately 5.3 times that of the control group, potentially linked to the activation of effector-triggered immunity (ETI). Although the expression of this gene decreased at 24 h, it began to rise gradually after 48 h. The high expression of *SbiRTX2783.06G018400* during the early stages of aphid infestation may be related to the initial resistance response, inhibiting further feeding by aphids. As aphid populations increased, the expression of this gene rose again, indicating its critical role in sustained aphid resistance. Additionally, the expression of other candidate genes also increased, albeit less dramatically, which aligns closely with the pattern-triggered immunity (PTI) mechanism.

Discussion

Sorghum is a vital crop for food and feed, and its resistance to aphids is essential for enhancing yield and minimizing pesticide usage. Investigating QTLs associated with aphid resistance and their underlying genes is crucial for understanding the mechanisms of resistance in sorghum and for breeding initiatives [33]. Zhang et al. employed whole-genome resequencing and transcriptome analysis of Recombinant inbred line (RIL)

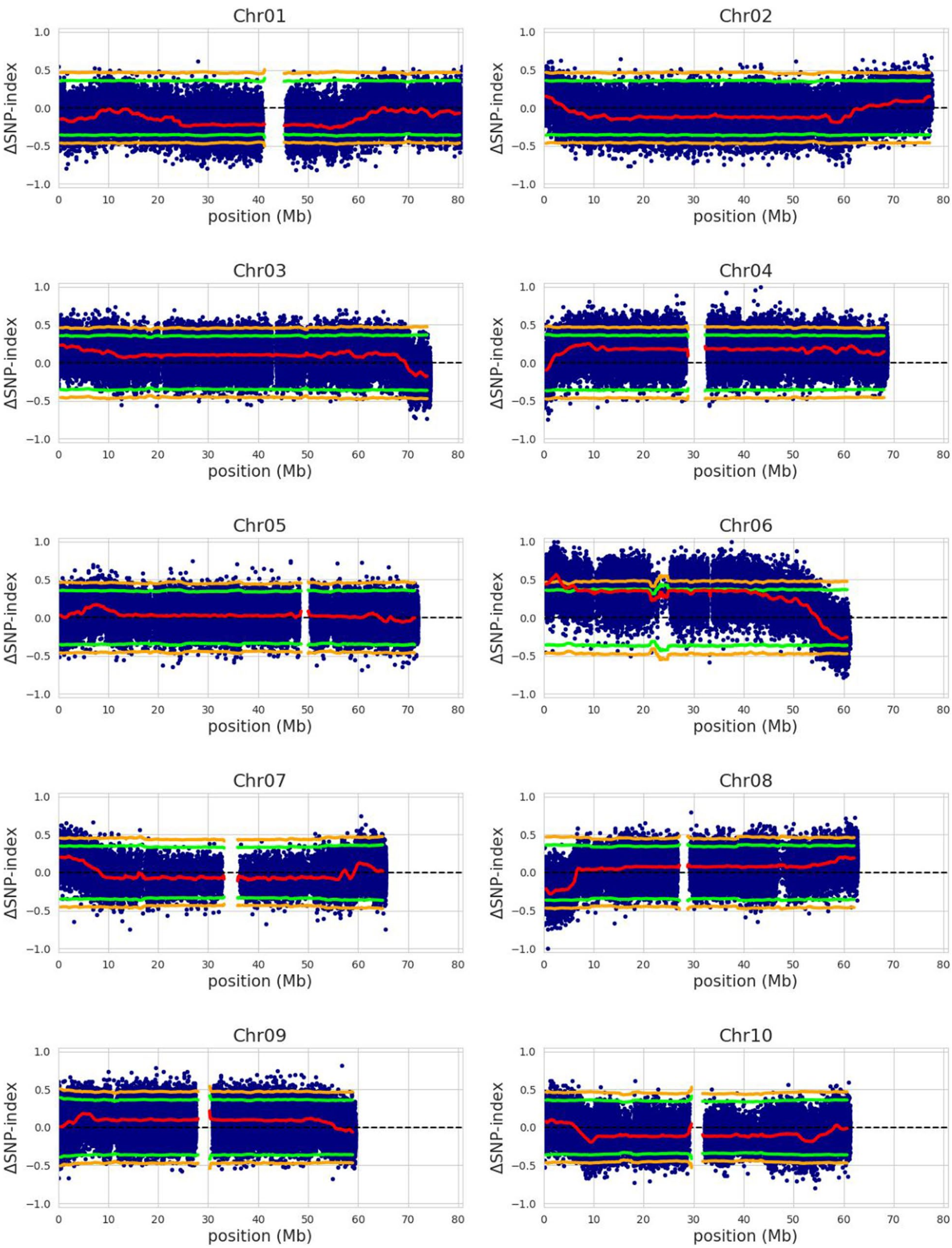

Fig. 4 Presents the distribution of the ΔSNP-index across the sorghum chromosomes. The blue scatter points indicate ΔSNP-index values at various positions along the chromosomes, while the red line represents a smoothed ΔSNP-index curve used to identify confidence intervals within the genome. The green and yellow curves correspond to 95% and 99% confidence thresholds, respectively, highlighting regions of significant association with the target trait

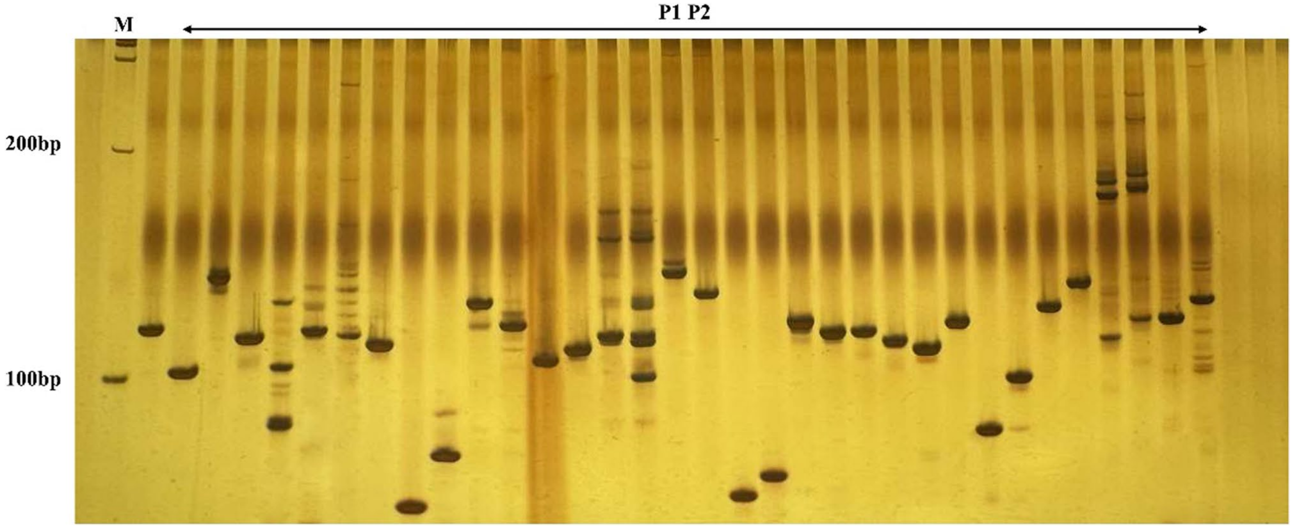

Fig. 5 Amplification results of InDel markers between parents (10% Polyacrylamide Gel Electrophoresis). P1 represents the aphid-resistant parent IS23514; P2 represents the aphid-susceptible parent IS23992; and M denotes the Marker

populations derived from 407B and 7B to identify QTLs for resistance to sorghum aphids. They discovered four aphid resistance QTLs on chromosome 6, designated as qtlMs-6.1, qtlMs-6.2, qtlMs-6.3, and qtlMs-6.4. Notably, one QTL (Chr6:2686447 C > G) included an SNP marker that was validated across diverse populations [20]. In the SC112-14 line, several important QTL loci were identified, with the most significant QTL located approximately 8–10 cM upstream of the known aphid resistance gene *RMES1*, within the 2.32–2.41 Mb region on chromosome 6. Given that *RMES1* is widely studied and utilized as an aphid resistance gene in sorghum, this suggests that the aphid resistance loci in the SC112-14 line may have some connection or synergistic effect with the resistance mechanisms of *RMES1* [34]. Moreover, recent studies have identified that the *RMES2* gene, located on chromosome 9 of sorghum at the 61.5–62.7 Mb region, also contributes to aphid resistance and is more commonly found in sorghum populations. *RMES2* functions synergistically with RMES1, further enhancing the plant's defensive mechanisms [35]. In this study, a QTL associated with aphid resistance was identified, located in a 124 kb region on sorghum chromosome 6, downstream of the *RMES1* gene and partially overlapping with it. This suggests that the region may be an important hotspot for aphid resistance traits. However, variations in the effect size of QTLs and candidate genes across different studies may be influenced by factors such as the genetic background of the population, phenotypic scoring criteria, and the aphid species used. Future research could further validate the broad applicability of this region by utilizing more sorghum germplasm resources and different aphid biotypes.

Pesticides are currently prevalent in agriculture, leading to significant disruptions in environmental health and ecosystems [36]. Investigating the mechanisms of sorghum aphid resistance has become essential for mitigating the adverse effects of pesticide misuse. As plants evolve and share genetic resources, they have developed various strategies to defend against pathogens and pests, primarily through R genes. Research by J. Calle Garcia et al. indicated that most R genes are proteins characterized by nucleotide-binding site (NBS) and leucine-rich repeat (LRR) domains, with the NBS-LRR gene family being the largest group of plant disease resistance genes [13, 37]. The proteins encoded by these genes typically participate in the recognition of pathogens and the ensuing defense responses in plants [38]. Notably, our study identified that out of the ten candidate genes, seven included the NB-ARC protein domain, categorizing them as R genes [39]. The NB-ARC domain, a fundamental component of plant resistance genes, has been thoroughly investigated for its role in pathogen recognition and signal transduction [40]. The seven candidate genes identified in this study may play a crucial role in inhibiting aphid feeding behavior by triggering downstream defense responses, such as hypersensitive reactions and programmed cell death [41]. The *Meu-1* gene was the first NBS-LRR resistance gene cloned from tomatoes [42], while the *Vat* gene, another NBS-LRR gene, confers resistance to cotton aphids (*Aphis gossypii*) [43]. Recently, Zhang et al. identified two sorghum aphid R genes, *RLK-1* (*Sobic.004G099700*)

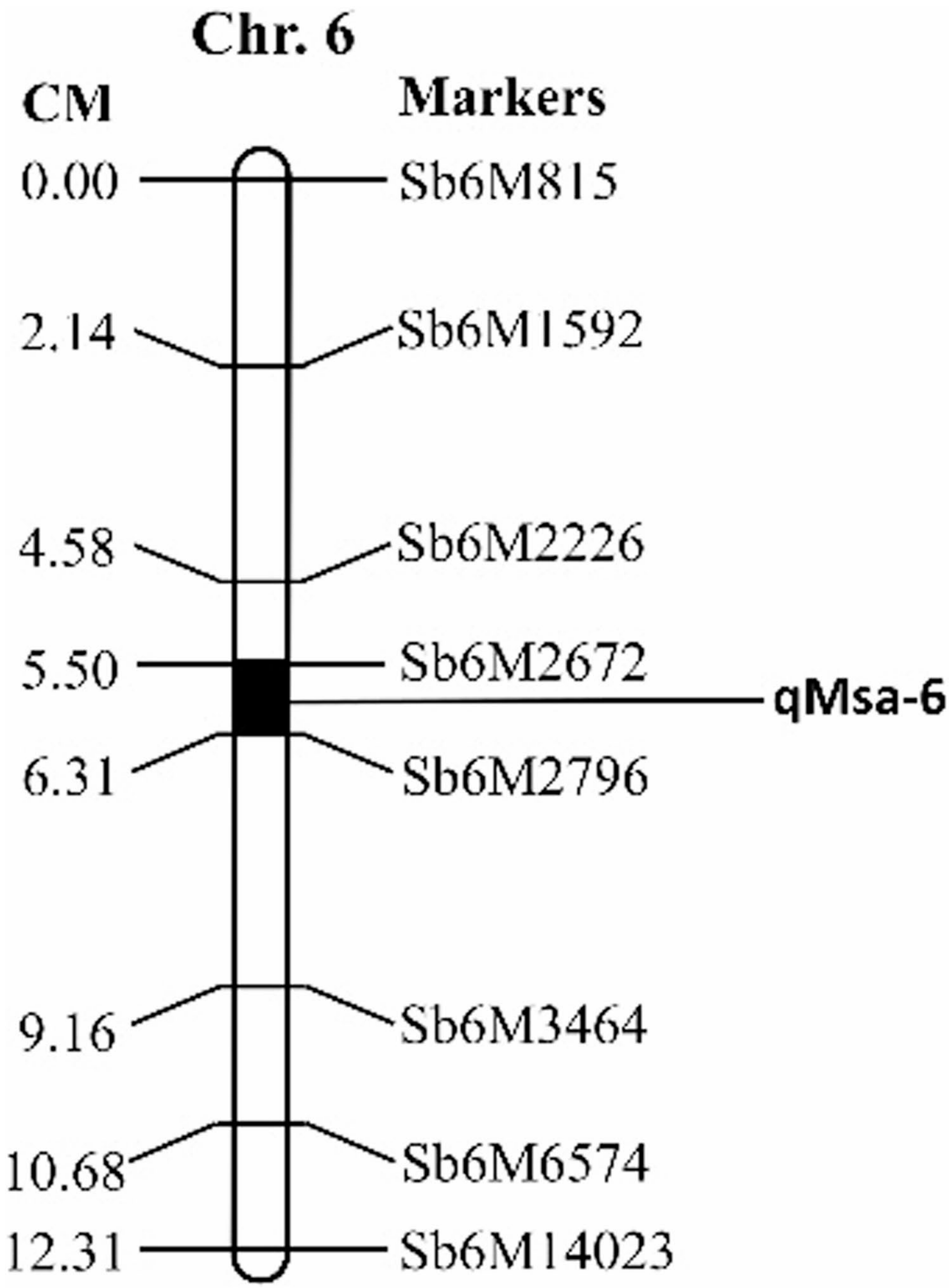

Fig. 6 Genetic linkage map of the sorghum aphid resistance QTL (qMsa-6) in the fine mapping interval on chromosome 6

and *NBS-1* (Sobic.003G325100), through whole-genome analysis [37], It is noteworthy that in our study, the strong induction of the *SbiRTX2783.06G018400* gene may be linked to the activation of ETI. ETI is a rapid and intense immune response produced by plants in response to specific pathogens, often accompanied by programmed cell death, which restricts the pathogen's spread at the infection site. This gene showed a highly sensitive response during aphid feeding, with significantly upregulated expression, indicating its crucial role in sorghum's aphid resistance. Other genes also exhibited increased expression, albeit to a lesser extent, potentially due to stress from rising aphid populations. This increase in expression aligns with the PTI mechanism, a more prolonged and low-intensity defense response that can slow pathogen spread but typically cannot fully prevent invasion.

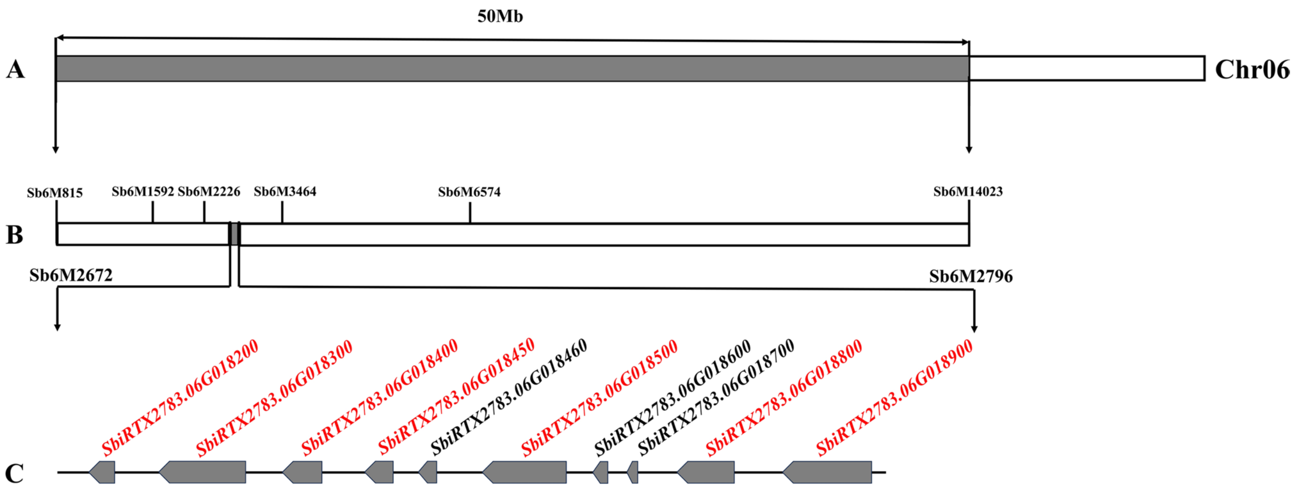

Fig. 7 Preliminary and fine mapping intervals. **A** show the preliminary mapping interval in gray based on the ΔSNP-Index. **B** displays the fine mapping interval marked in gray, defined by InDel markers. **C** highlights the candidate genes within the fine mapping interval, with the red genes containing the NB-ARC domain

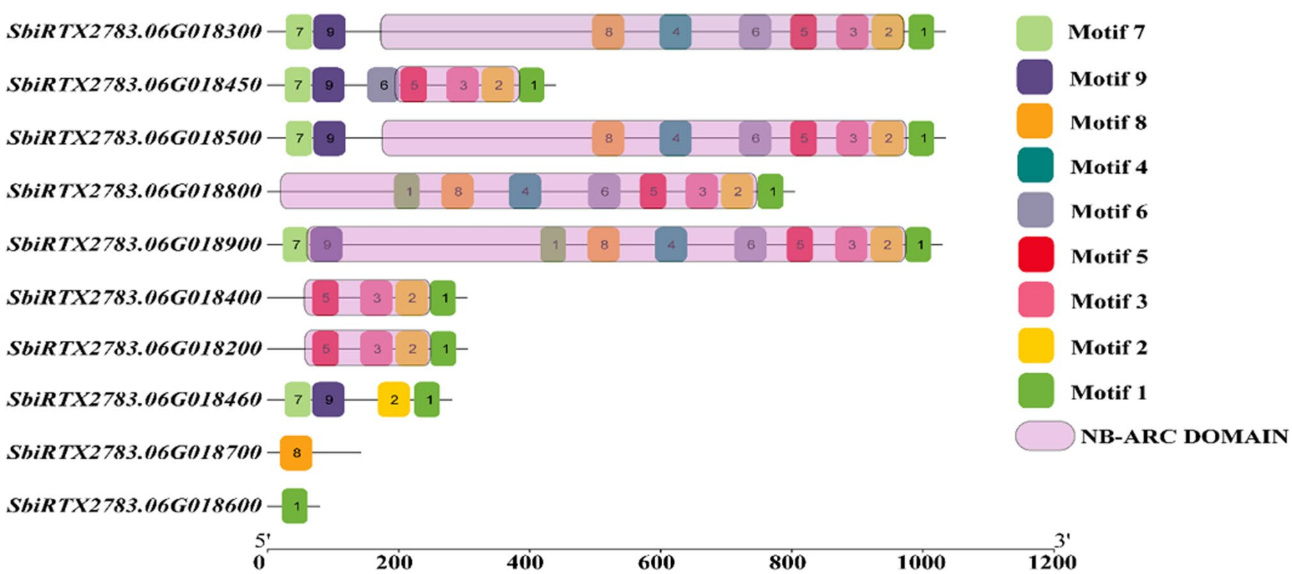

Fig. 8 Analysis of conserved domains and protein structures of candidate genes within the mapping interval in sorghum

Therefore, we propose that *SbiRTX2783.06G018400* may work in conjunction with the other six genes to form a multi-layered plant immune defense system [44–46].

Currently, sorghum aphid resistance remains a hot research topic, and the precise localization of aphid resistance QTLs in sorghum is still limited. This increases the urgent need for new aphid resistance QTLs and resistance genes. This study presents a novel QTL (qMsa-6) and *SbiRTX2783.06G018400* has been proposed as a candidate gene for aphid resistance due to its characteristics aligning with effector-triggered immunity (ETI). This discovery lays a solid molecular foundation for aphid-resistance breeding in sorghum. Incorporating these candidate genes in MAS can accelerate the breeding of aphid-resistant sorghum varieties. Moreover, with advancements in gene editing technologies, these resistance genes can be directly improved to develop more efficient and durable aphid-resistant varieties, offering a sustainable solution for sorghum production.

Conclusion

In this study, BSA-seq combined with InDel marker technology was employed to precisely map a QTL for aphid resistance on chromosome 6 of sorghum, designated as qMsa-6. This QTL spans a 124 kb interval, partially overlapping with the previously reported aphid resistance gene *RMES1*, but with higher mapping resolution. By

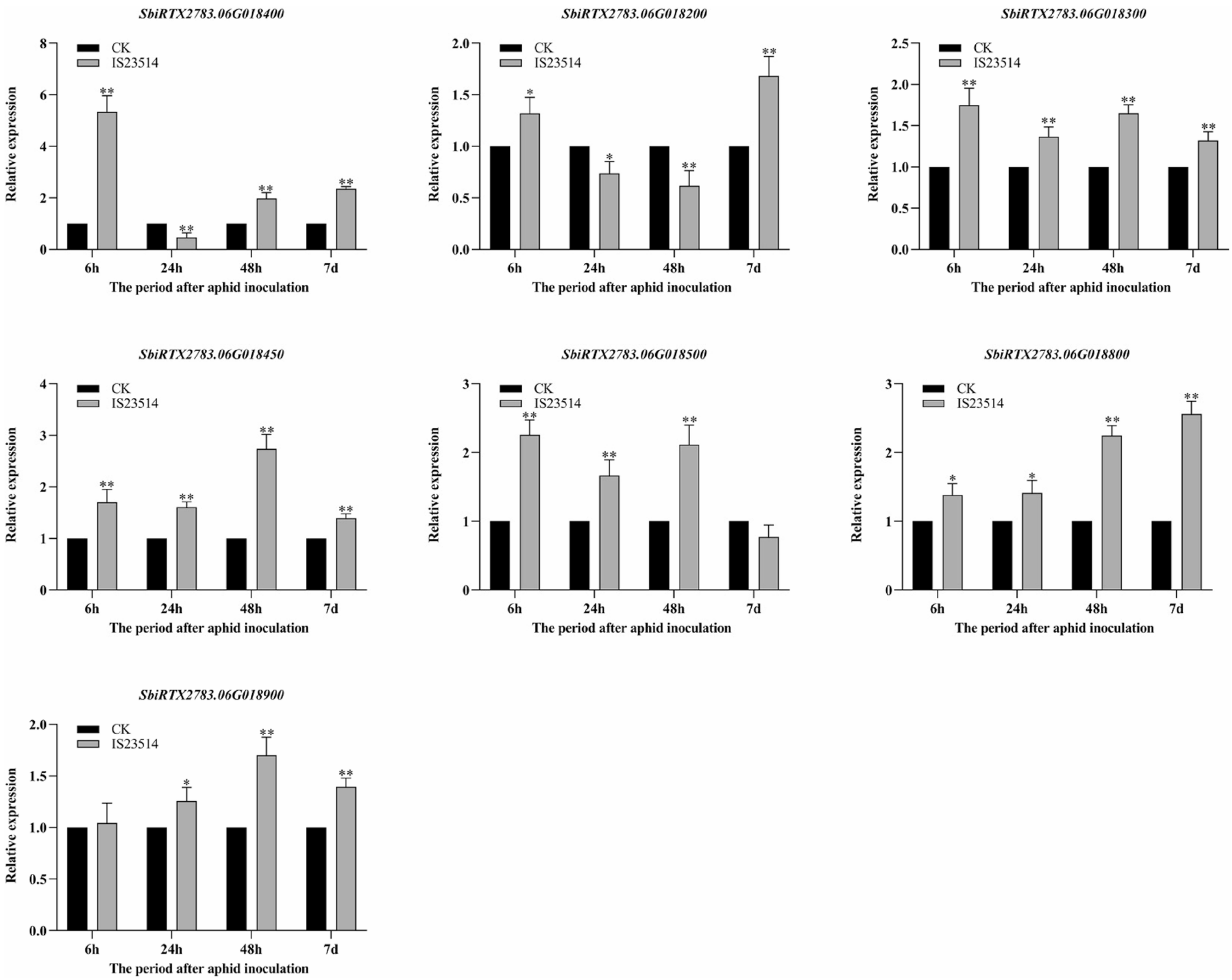

Fig. 9 Expression patterns of candidate genes after aphid invasion. The invasion time points were set at four early stages: 0, 6, 24, and 48 h post-invasion, with 0 h serving as the early CK (Control Group). A later time point was set at 7 days post-invasion, with aphid-free plants at 7 days used as the CK (Control Group). Significant differences are indicated by (** = $p < 0.01$, * = $p < 0.05$)

comparing differentially expressed genes (DEGs) in the candidate region between resistant and susceptible sorghum lines, ten previously uncharacterized candidate genes were initially identified. Subsequent bioinformatic analyses further narrowed this list to seven resistance (R) genes containing NB-ARC domains, which were selected as the final candidates. qRT-PCR expression profiling revealed that these NB-ARC domain-containing genes exhibited higher expression levels in aphid-resistant sorghum, with *SbiRTX2783.06G018400* showing significant upregulation upon aphid infestation, suggesting a key role in the immune response of sorghum to aphids. Although the specific functions and regulatory mechanisms of these candidate genes remain to be elucidated, this study provides important theoretical insights and practical references for fine-mapping of aphid resistance QTLs, functional gene cloning, and the molecular breeding of aphid-resistant sorghum.

Abbreviations

BSA-seq	Bulked segregant analysis sequencing
QTL	Quantitative trait locus
MAS	Marker-assisted selection
F_1	First filial generation
F_2	Second filial generation
InDel	Insertion-Deletion
qPCR	Quantitative real-time polymerase chain reaction
SSR	Simple sequence repeat
SNP	Single nucleotide polymorphism
PCR	Polymerase chain reaction
PAGE	Polyacrylamide gel electrophoresis
NBS	Nucleotide-binding site
LRR	Leucine-rich repeat
NB-ARC	Nucleotide-binding domain shared by APAF-1, R proteins and CED-4
ETI	Effector-triggered immunity
PTI	Pattern-triggered immunity
RIL	Recombinant inbred line
KASP	Kompetitive allele specific PCR
SCAR	Sequence characterized amplified region
R-pool	Resistant pool
S-pool	Susceptible pool
Mb	Megabase

kb Kilobase
cM Centimorgan

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12870-025-07028-1.

Supplementary Material 1

Acknowledgements
Not applicable.

Authors' contributions
Author Contributions Conceptualization, J.L. and J.D.; methodology, J.L.; software, K.Z. and Q.S; validation, T.W., M.G., Q.S. and Y.L.; formal analysis, K.Z., J.L.; investigation, K.Z., J.D. and Y.L.; resources, J.L. and J.D.; data curation, K.Z., L.W., Y.W., T.W. and Q.S.; writing—original draft preparation, K.Z., Y.L., M.G., T.W., Q.S., L.W., J.L., J.D., Y.W. and D.W.; writing—review and editing, K.Z., J.D., Q.S., J.L. and M.G.; visualization, K.Z., Q.S. and Y.L.; supervision, D.W. and J.D.; project administration, J.D. and K.Z.; funding acquisition, J.L., J.D. and D.W. All authors have read and agreed to the published version of the manuscript.

Funding
This work was supported by the National Natural Science Foundation of China (32372134), Key Discipline Construction Fund for Crop Science of Anhui Science and Technology University (No. XK-XJGF001), Natural Science Foundation of Education Department of Anhui Province (2023AH051852), and supported by Anhui Province International Joint Research Center of Forage Bio-breeding (No. AHIJRCFB202303).

Data availability
The raw BSA-seq reads generated and analysed during the current study have been deposited in the CNGB Sequence Archive (CNSA) of the China National GeneBank DataBase (CNGBdb) under accession number CNP0007077 (https://db.cngb.org/search/?q=CNP0007077).

Declarations

Ethics approval and consent to participate
Experimental research and field studies on plants and aphids, including the collection of plant material and aphid rearing, complied with relevant institutional, national, and international guidelines and legislation.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 18 April 2025 / Accepted: 7 July 2025
Published online: 16 August 2025

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